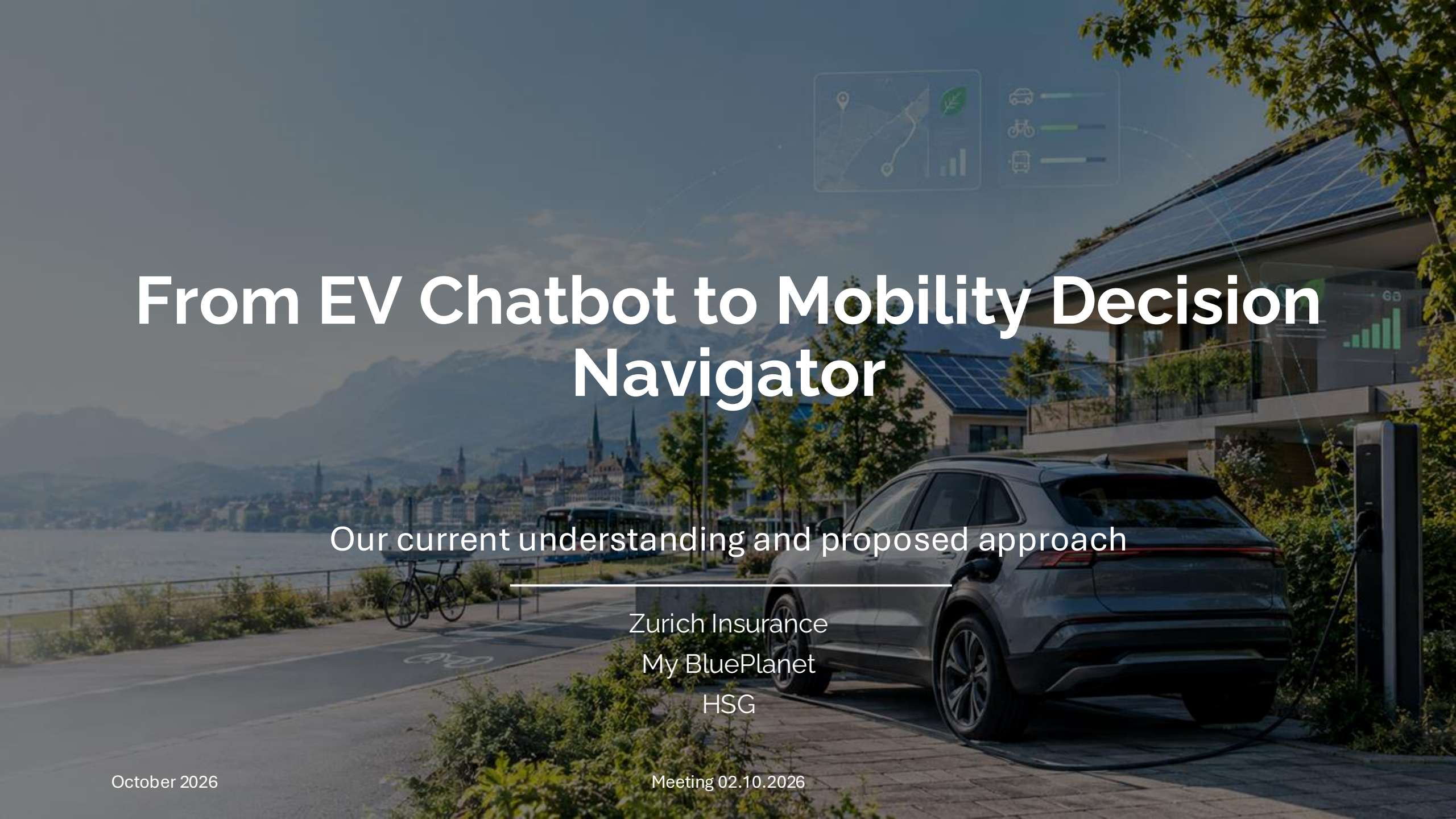


The background image shows a modern, two-story house with a large array of solar panels on the roof. A dark-colored electric car is parked in front of the house, plugged into a charging station. In the background, a city with a prominent church spire is visible across a body of water, with mountains in the distance. The scene is set during the day with soft lighting.

From EV Chatbot to Mobility Decision Navigator

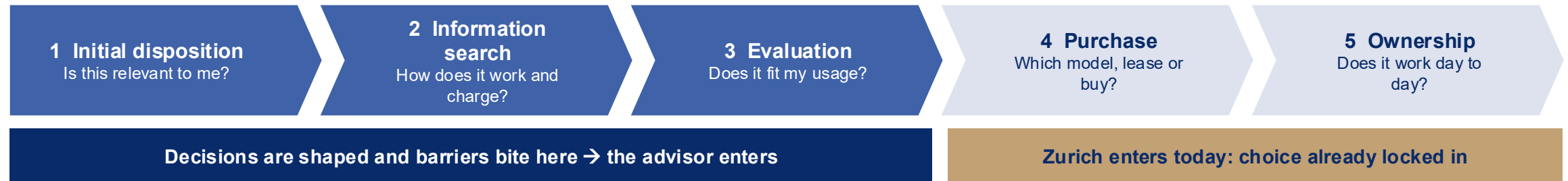
Our current understanding and proposed approach

Zurich Insurance
My BluePlanet
HSG

Problem

Zurich meets customers after the vehicle decision is made, exactly where genuine barriers decide the outcome

Swiss BEV purchase journey: five phases (INFRAS, 2026)



Three findings shape the solution design

#1

Charging access is the decisive factor when users weigh up a BEV. No private charging is rated a critical barrier.

Source: INFRAS (2026)

61%

of the Swiss population rents. Rented and multi-party housing is the hardest charging problem to solve locally.

Source: mobilservice.ch

>80%

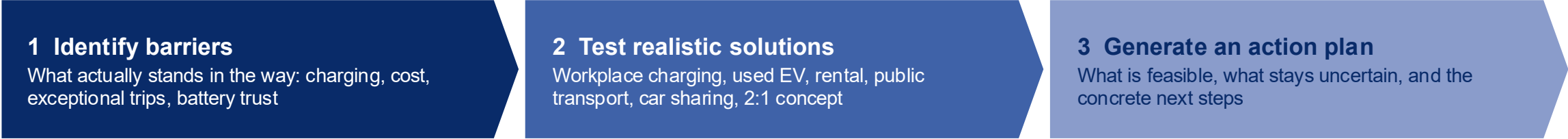
of BEV owners would not go back to combustion (84% satisfied), yet public scepticism persists: a perception gap a neutral advisor can close.

Source: Roland Berger 2025, ADAC 2025, cited in INFRAS

Implication: influence must move upstream to phases 1- 3, with an advisor that treats barriers as legitimate concerns to solve, not objections to overcome.

Solution

We propose a Mobility Decision Navigator: a neutral, evidence-based service, not an EV recommendation chatbot



	Typical EV recommendation tool	Mobility Decision Navigator
Goal	Convince users to buy a BEV	Help users find what is realistic, including not buying
Opening question	“Which EV do you want?”	“Help me understand your lifestyle, needs, constraints and goals”
Outcomes	One: a car model	Several equal outcomes, from BEV now to keep the car, used EV, public transport, car sharing or the 2:1 concept
Vehicle sizing	Largest car “just in case”	Right-size for 80–90% of daily use; rental or 2:1 for the rest
Decision logic	Free-form LLM conversation	Transparent rules and evidence; AI assists but does not decide
Result	Model shortlist or readiness score	Economic assessment, personalized recommendation, barriers & solution map, action plan

Success metric: every user leaves with a clear next step, even when that step is not buying a BEV.

Solution – 10 personas

Personas steer the conversation: barriers and transparent rules, not AI improvisation, drive the recommendation

~10 Swiss mobility personas¹

1 Urban Renter	2 Family Homeowner	3 Rural / Alpine Multi-Car	4 Long-Distance Commuter	5 Company-Car User
6 Cost-Conscious Driver	7 Sustainability-Oriented User	8 Skeptical Traditionalist	9 Occasional Urban Household	10 Multi-Vehicle 2:1 Household

1 Persona	2 Barriers	3 Solutions	4 Action plan
FRAMEWORK Adapts questions, tone and explanations. Users may match several personas, we show the overlap, not one forced label.	FRAMEWORK Every question traces to a barrier: charging, affordability, trust, infrastructure, practicality, other	FRAMEWORK Shows whether a BEV is feasible, under which conditions (e.g. workplace or shared-garage charging, used EV, 2:1 concept) and how it compares economically with keeping the current car.	FRAMEWORK Concrete transition plan with prioritized next steps such as contacting a landlord about charging, requesting workplace charging, obtaining a battery-health certificate, booking a 48–72 hour test drive, or evaluating alternative mobility options before making a purchase decision
EXAMPLE: URBAN RENTER City apartment, no assigned parking with charging access.	EXAMPLE: URBAN RENTER Critical: home charging access. Relevant: public charging, upfront cost.	EXAMPLE: URBAN RENTER Ask landlord, check workplace charging, map public chargers nearby; car sharing in the meantime.	EXAMPLE: URBAN RENTER BEV after solving charging, or car sharing / public transport if no path in ~12 months. Landlord letter, revisit date.

Design rule: the same decision logic applies to every user, personas change the path and the language, not the standard of evidence.

1: Generated with AI only for explanation purposes. The next step would be to conduct interviews or use relevant studies such as [this study](#)

Solution – n8n workflows

The architecture separates interface, decision rules and evidence – AI supports, it does not decide

USER LAYER

Website interface

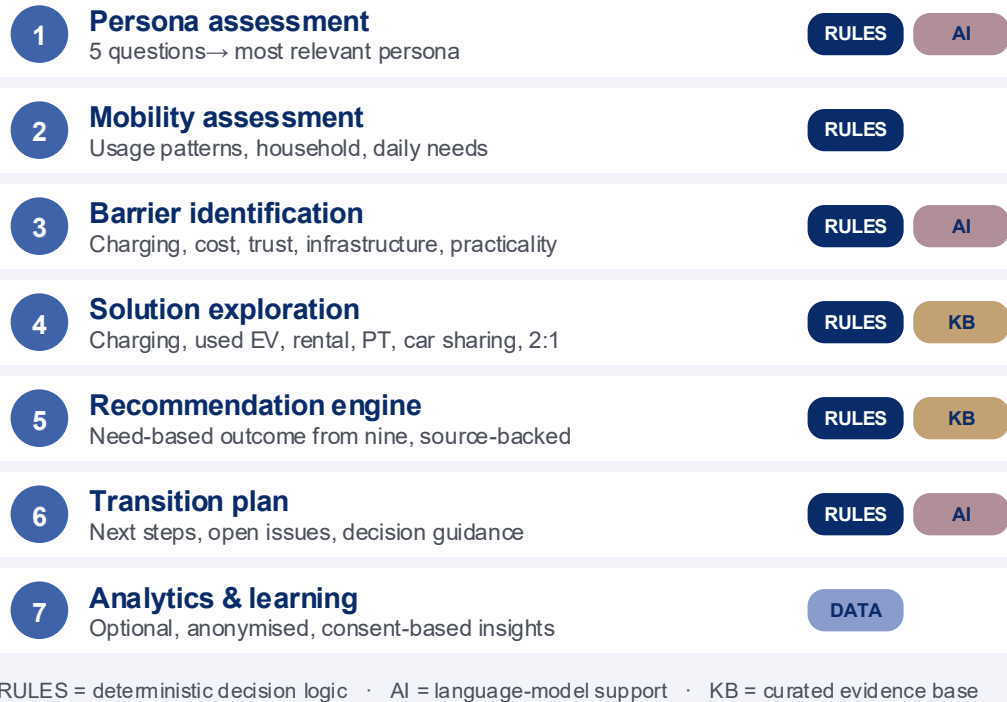
- Guided, button-first chat, minimal typing
- More detailed questions linked to AHA moments, e.g. nearby charging options
- Consent and “not a sales tool” notice upfront
- No registration by default

Output

Dashboard “Your BEV Check”

For more details, check the file:
“BEV Navigator – User Journey Concept”

ORCHESTRATION IN n8n · 7 WORKFLOWS



SUPPORT SERVICES

AI model(s)

Free-text barrier detection, persona support, plain-language explanations. Bounded by rules; no final decision.

Knowledge base (RAG)

Curated trusted sources with citations; facts, estimates and assumptions labelled.

Analytics store

Anonymised barrier and behaviour insights; optional link to the Climate Action App.

Design principles

Guided, not typed

Buttons keep users engaged

AHA moments

Surface solutions users don't know

Source-backed

RAG with citations, fewer hallucinations

Facts vs. estimates

Assumptions and limits explicit

Privacy-first

Minimal data, in-line purpose

Solution – Knowledge base

Proposed: curated, citable sources, official Swiss data first, partner data where there are gaps

Area	Proposed source	Publisher	Status
Barriers & priorities	Barrier analysis for the Canton of Zurich (2026) · Zurich briefing (2026)	INFRAS / AWEL · Zurich	In hand
Charging points near the user	ich-tanke-strom open data: 19,479 points, 41 operators (largest networks only)	BFE	Tested
Electricity price by location	Tariffs per municipality, grid operator and household profile (open data)	EICom	Tested
Car classes, consumption, running costs	Verbrauchskatalog: all new cars, consumption, CO2, TCS cost model	BFE / TCS	Candidate
Typical driving (personas)	Mikrozensus Mobilität und Verkehr 2021 (pandemic year)	BFS / ARE	Candidate
Right to charge	Federal Council bill in consultation until 12 Oct 2026: shown as a proposal, not law	Federal Council	Checked
Rental & used-car prices	Renault dealership	Partner	To request
Mobile DC charging	Specifications and costs	Designwerk	To request

No source yet: public charging tariffs, cantonal vehicle tax · BFE data: commercial use may need BFE permission · Z-volt data

Our proposal: official Swiss open data first, partner data only where there are gaps.

Appendix 1: BEV Navigator – User Journey Concept

